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8

Wind Energy

8.1 Learning Outcomes

After actively engaging with the material in this chapter, you should be able to

1. Explain the function of a wind turbine in converting the kinetic energy of a wind stream into rotational mechanical energy;
2. Define the terms cut-in, rated, and shutdown speeds, and the rated power of a wind turbine;
3. Derive the Betz limit on efficiency of a wind turbine;
4. Derive expressions for the input and output powers of a wind turbine;
5. Calculate the annual energy output from a given turbine using its power wind speed curve and the wind speed distribution for its location;
6. Discuss the main factors that determine the performance characteristics of a wind turbine;
7. Discuss the advantages and disadvantages of wind energy.

8.2 Overview

This chapter provides an introduction to the subject of wind energy. It presents a simple method for deriving mathematical expressions for the input and output powers of a wind turbine. This is followed by a discussion of vertical

and horizontal wind turbines and their advantages and disadvantages. The Betz limit, which sets an upper bound on the theoretical efficiency of a wind energy conversion system, is also derived. Calculations of the annual energy output from a wind turbine using its power-speed curve and the distribution of the annual wind speed for its location is explained. Finally, several problems are presented at the end of the chapter to consolidate understanding of the material.

8.3 Introduction

A moving fluid, such as wind, has mass and velocity and therefore has kinetic energy that can be harvested and converted into a usable form of energy. In this chapter, we focus on the technology of harvesting wind energy and converting it into electrical energy, which is the most convenient form of energy. A wind turbine extracts the kinetic energy in a moving wind stream and converts it into rotational mechanical energy, which rotates an electric generator. This assembly of a wind turbine and a generator is a wind turbine-generator (WTG) set; however, we normally refer to it simply as a wind turbine.

Wind energy is another manifestation of solar energy. It is solar energy that heats different regions of the earth's surface to different temperatures, thus producing wind. The annual solar radiation received per unit area of the earth's surface in tropical regions is far more than that received by regions at higher latitudes. Due to the curvature of the earth, its surface becomes more slanted with increasing latitudes. Consequently, sunrays must travel longer distances to reach higher latitudes, thus losing more energy to the atmosphere before they reach the earth's surface. Variations in the temperature of the earth's surface between different regions results in variations in atmospheric pressure, which gives rise to the movement of air molecules, thus producing wind [1]. Wind energy is one of the most ancient forms of energy known to humans. It has been used to propel sailboats and grind grains for thousands of years. However, its use to generate electricity is relatively new; the first wind turbine to produce electricity was built in Scotland by James Blyth in 1887, and around the same time, a 12-kW fully functional turbine was built in the United States by Charles Brush [2].

8.4 Turbine Blades

The main part a wind turbine is the array of blades, typically two or three, fitted equally spaced around a shaft, as shown in Figure 8.1.

The blades are shaped so that the cross section of each blade, also included in Figure 8.1, is an airfoil all the way from the trailing to the leading edge. Because of this airfoil shape, a lift force is generated when wind blows over a blade,

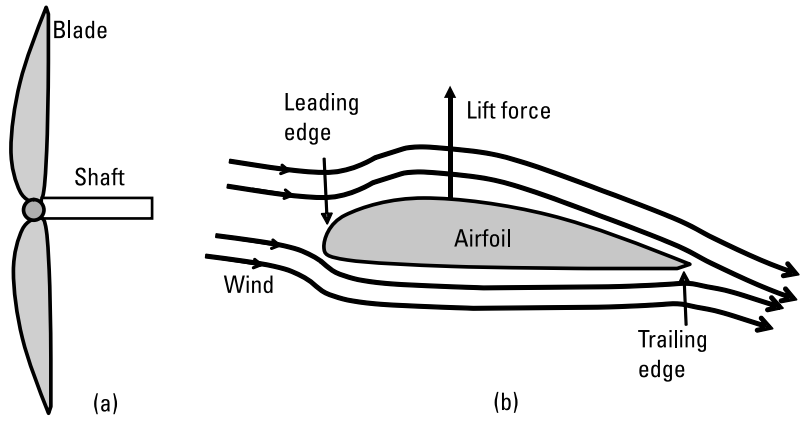


Figure 8.1 (a) Two-blade turbine, and (b) production of lift force.

which causes the blade to rotate. As the blades rotate, the kinetic energy in the wind is transferred to the turbine shaft (i.e., the rotor) as a rotational mechanical energy [3].

8.5 Types of Wind Turbines

One method of classifying wind turbines is in reference to the axis around which the turbine blades rotate. Consequently, a turbine is classified as horizontal axis wind turbine (HAWT) or vertical axis wind turbine (VAWT). In HAWTs, the axis of rotation is in line with the direction of wind, whereas in VAWTs the rotation axis is perpendicular to the wind direction. Wind turbines range in size from very small machines that produce few tens or hundreds of watts to very large turbines producing several megawatts. In general, in the context of wind electricity generation, unless otherwise stated, a wind turbine also includes an electric generator, a breaking system, a gearbox, and all the associated control devices.

8.6 Horizontal Axis Wind Turbines

HAWTs, which are used for electricity generation, typically have either two or three blades. The structure of a typical HAWT system is shown in Figure 8.2. The main components of the system are enclosed in the nacelle. The wind speed is normally much lower than the speed required to run the generator to generate electricity with the required frequency, 50 Hz in Europe and 60 Hz in the United States. Therefore, a gear box is used to step up the turbine speed before it is used to spin the generator. A wind turbine may also include an an-

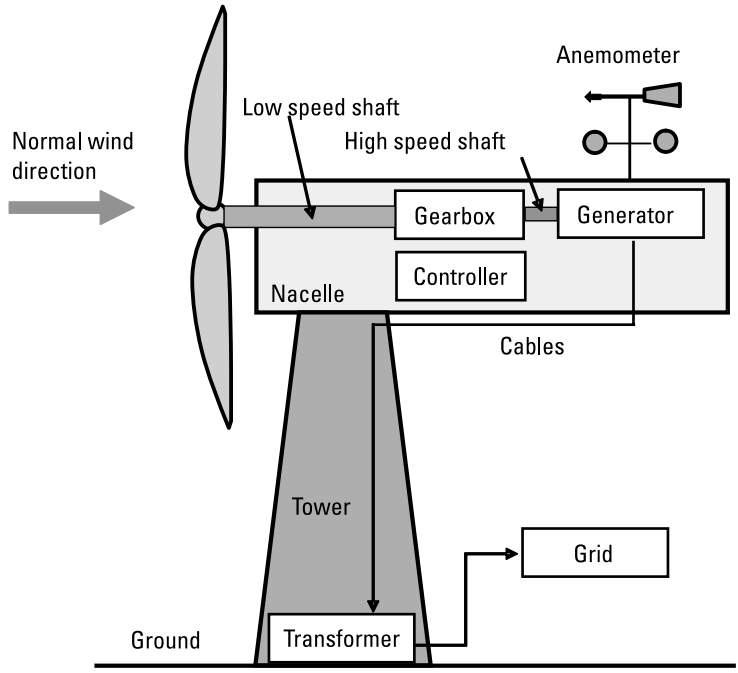


Figure 8.2 A typical HAWT and main components.

emometer, which measures the direction and speed of wind. The anemometer feeds the wind direction measurement as an electrical signal to the controller, which will send a signal to the yawing mechanism to rotate the nacelle so that the turbine will always be aligned to the wind direction for maximum efficiency of wind energy extraction.

This efficiency of wind energy conversion also depends on the angle of the wind-relative velocity known as the angle of attack, shown in Figure 8.3, which changes with the wind speed. Hence, a control signal representing wind speed is also sent to the controller to tilt the blades to keep them aligned for maximum energy extraction. Although not shown in the diagram, a wind turbine also includes a braking system for stopping the turbine in the event of an excessive wind speed that can damage the turbine if it is allowed to spin. The transformer is used to step up the output voltage of the generator to a level suitable for transmission and integration with the existing utility grid.

An important issue in the design of a horizontal wind turbine is whether to use an upwind or a downwind turbine, as shown in Figure 8.4. A downwind turbine has the advantage of letting the wind itself control the yaw, so the turbine automatically aligns itself correctly with respect to wind direction. However, there is a major problem with this design due to the wind shadowing effects of the tower. This is because every time a blade swings behind the tower,

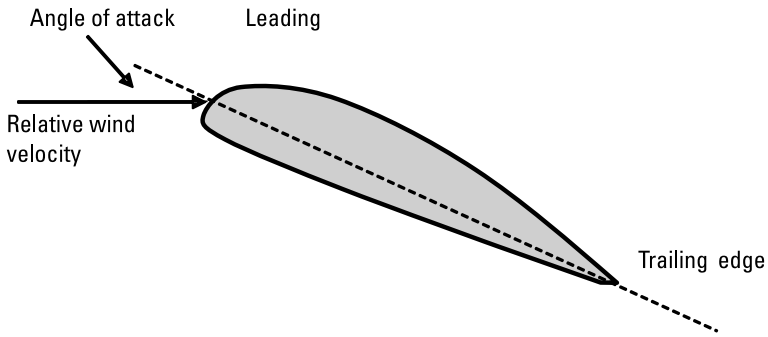


Figure 8.3 The angle of attack of the relative velocity.

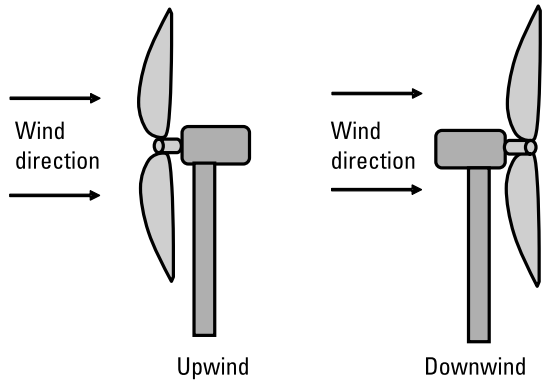


Figure 8.4 (a) Upwind and (b) downwind turbines.

it encounters a brief period of reduced wind. This causes the blade to flex, which increases the risk of blade failure, generates noise, and reduces the power output of the turbine. Upwind turbines, on the other hand, require complex yaw control systems to keep the blades facing the direction of the wind but are much quieter and more efficient. Currently, most electricity generating wind turbines in use are of the upwind type [4].

8.7 Vertical Axis Wind Turbines

Three different types of VAWTs are shown in Figure 8.5. The Savonius turbine was invented by the Finnish architect Laurentius Savonius in 1922. It consists of two semicylinders or cups. When the open cup is facing the direction of the wind, the force on it will be much larger than the force at the back of the other cup because the surface of the latter is rounded. This produces a dragging force

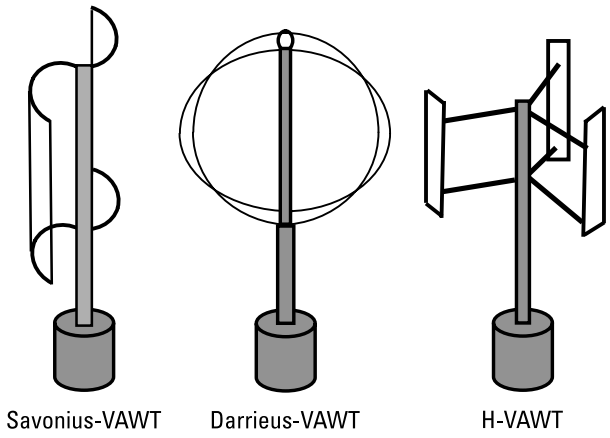


Figure 8.5 Three different types of VAWTs.

that will rotate the rotor. The Darrieus vertical axis turbine, named after its inventor, consists of a number of curved airfoil blades, thus it is a lift type of turbine. A Darrieus turbine runs much faster than a Savonius turbine; in fact, it can run at much higher speed than the speed of wind striking it, but the torque it develops is less than that of a Savonius turbine. The high-speed capability makes a Darrieus turbine suited for electricity generation. The H-type VAWT, also shown in Figure 8.5, has simple construction because it uses straight blades attached to a shaft using struts, which makes it easier to manufacture, and it is cheaper than the Darrieus turbine. In addition, the H-type turbine has better aerodynamic performance than the Darrieus type [5].

The main advantage of VAWTs is that they can harvest winds from any direction without the need for complex yaw control to continually reposition the rotor whenever the wind direction changes. The second advantage of VAWTs is that the heavy system components, such as the gearbox and generator, can be located on the ground, alleviating the need for structurally strong nacelle and strong tower as in the case with HAWT, which makes these components easily accessible for maintenance. The tower can be even made lighter and thus cheaper, particularly when guy wires are used. However, despite these advantages, they have found little commercial success to date due to several disadvantages [4]. The principle disadvantage of VAWTs is the fact that the blades are close to the ground. This is a major drawback because close-to-ground wind speeds are slower and since the power in the wind depends on the cube of the wind speed, this places considerable limitation on the output power of these turbines. Further, at excessively high wind speeds, when it becomes necessary to control the speed of a turbine to protect the generator, it is difficult to make the vertical blades spill the wind as is the case with pitch-controlled blades on the HAWT. In addition, wind is more turbulent the closer to the ground it is,

which increases stress on the VAWT [6]. Another fundamental design decision for wind turbines relates to the number of rotating blades. Wind turbines with many blades operate with much lower rotational speed than those with fewer blades. As the revolution per minute (rpm) of the turbine increases, the turbulence caused by one blade affects the efficiency of the blade that follows. With fewer blades, the turbine can spin faster before this interference becomes excessive. A faster spinning shaft means that generators can be physically smaller in size for a given power rating. Most modern European wind turbines have three rotor blades, while American ones tend to have just two. Three-bladed turbines tend to be quieter; however, the third blade increases the weight and cost of the turbine by a considerable margin [4].

8.8 Power in the Wind

Since wind generates electricity, it is important that we can estimate the amount of power in a wind stream. The wind is a continuously moving fluid; hence, it is delivering energy continuously, and therefore we really need to determine the rate at which this energy is delivered by the wind (i.e., the power in watts). In order to evaluate the rate at which kinetic energy is delivered by wind, we need to know the mass of air reaching the turbine per second (i.e., the mass flow rate (m dot) $\dot{m}$(kg/s)). This requires the density (ρ) of air, which depends on temperature and altitude. Dry air at normal pressure at zero degrees Celsius has a density of 1.29 kg/m^{3}and at 15 degrees, its density is approximately 1.23 kg/m^3 [1]. Now let us determine how much energy is delivered by a wind stream to a turbine with radius R (m), as shown in Figure 8.6.

The kinetic energy of mass m (kg) of air traveling at speed of v(m/s) is

$$E_K = \frac{1}{2}mv^2 \tag{8.1}$$

The area swept by the turbine is

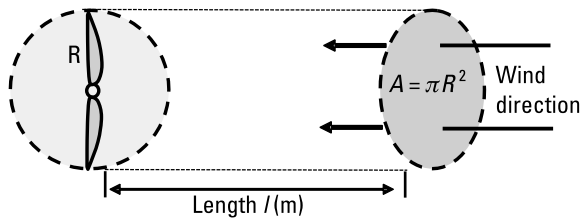


Figure 8.6 Determining the power delivered to a turbine.

$$A = \pi R^2 \tag{8.2}$$

Since the speed of wind is v(m/s), then in one second a cylinder of area A(m^2) and length v(m) will pass the turbine. Hence, the volume of air passing the blades per second is

$$V = Av \quad (\text{m}^3/\text{s}) \tag{8.3}$$

and the mass of air passing the turbine per second (i.e., the mass flow rate $\dot{m}$ (kg/s)) is

$$\dot{m} = \rho Av \quad (\text{kg/s}) \tag{8.4}$$

Hence, the kinetic energy delivered by the wind to the turbine per second (in J/s or W) (i.e., the power) is

$$P_w = \frac{1}{2}\rho Av^3 \quad (\text{J/s}) \tag{8.5}$$

It is clear from the above relationship that the power in the wind is proportional to the density of the air, the area through which the wind is passing, and the cube of the wind velocity. Hence, velocity is very important for power extraction. It is imperative at this point to appreciate that the wind velocity, in the above expression for power, is the undisturbed wind velocity (i.e., it is the velocity that is unaffected by the turbine). However, not all of this power is converted to useful rotational mechanical energy by the turbine; as we will soon see it only represents the power in the wind.

8.8.1 Effect of the Number of Blades

Consider a point on the tip of a blade on the rotor of a turbine, as shown in Figure 8.7. As the blade rotates, its tip sweeps out a circle whose radius is R (m) (i.e., the length of the blade). This corresponds to an angle of 360 degrees or 2π radians. The distance traveled by the tip is the circumference of the circle (i.e., $2\pi R$). When the tip rotates through an angle of only one radian (i.e., approximately 57.3°), the tip has traveled through a distance of R (m) (see Figure 8.7). Similarly, when the rotor turns through one complete revolution, the tip travels a distance equal to $2\pi R$ (m). If the angular speed of the rotor is ω radians per second (rad/s), then the actual tip speed U in m/s is simply R times the angular velocity

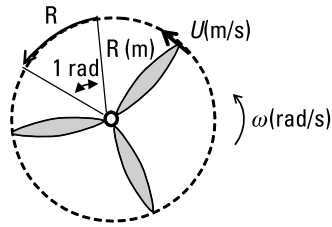


Figure 8.7 Angular and tip speeds.

$$U = \omega R \tag{8.6}$$

This is the tangential velocity of the rotor at the tip of the blades. Usually, the speed of a wind turbine is given in rpm. If the rotor is rotating at speed N (rpm), then it sweeps 2π radians every minute. Hence, its angular velocity in radians per second becomes

$$\omega = \frac{2\pi N}{60} \tag{8.7}$$

And the tip speed becomes

$$U = \frac{2\pi RN}{60} \tag{8.8}$$

By dividing the tip speed, U, by the undisturbed wind velocity, v(m/s) (i.e., the upstream of the rotor), we obtain a nondimensional ratio known as the tip speed ratio (TSR)

$$\text{TSR} = \frac{U}{v} \tag{8.9}$$

This ratio provides a useful measure against which the aerodynamic efficiency of a turbine can be assessed. The aerodynamic efficiency of a wind turbine is usually described by its power coefficient, which is defined as the ratio of mechanical power output from the turbine to the theoretical power in the wind. This quantity is given the symbol C_P and can be expressed as a decimal fraction or a percentage

$$C_p = \frac{\text{mechanical power output from the turbine}}{\text{theoretical power in the wind}} \tag{8.10}$$

When the power coefficient is plotted against TSR for a number of turbines, such plots provide an effective way to present the performance of a given turbine and to compare wind turbines with differing characteristics [4].

A wind turbine of a particular design can operate over a range of TSR, but will usually operate with its maximum efficiency at a particular (i.e., optimum) TSR. This occurs when the velocity of its blade tips is a particular multiple of the wind velocity. This optimum TSR is commonly denoted as TSR_{max} with the corresponding power coefficient being C_{Pmax}. The optimum TSR for a given wind turbine rotor depends on both the number of blades and the width of each blade.

In order to extract energy as efficiently as possible, the blades have to interact with as much of the wind passing as possible through the rotor's swept area. The blades of a multiblade wind turbine interact with all the wind at very low TSRs, whereas the blades of a turbine with few blades have to travel much faster to virtually fill up the swept area in order to interact with all the wind passing through. If the TSR is too low, some of the wind travels through the rotor swept area without interacting with the blades, whereas if the tip speed ratio is too high, the turbine offers too much resistance to the wind, so that some of the wind goes around it. A two-bladed wind turbine rotor with each blade the same width as those of a three-bladed rotor will have an optimum TSR one-third higher than that of a three-bladed rotor. Optimum TSRs for wind turbines with a small number of blades range between about 6 and 20 [4].

In theory, the more blades a wind turbine rotor has, the more efficient it is. However, when there is a large number of blades in a rotor, the flow becomes more disturbed, so that they aerodynamically interfere with each other. Therefore, multiblade wind turbines tend to be less efficient overall than those with few blades. Three-bladed rotors tend to be the most energy efficient, two-bladed rotors are slightly less efficient, and one-bladed rotors slightly less efficient still. Wind turbines with more blades can be generally expected to generate less aerodynamic noise as they operate at lower tip speeds than wind turbines with fewer blades. The mechanical power that a wind turbine extracts from the wind is the product of its angular velocity and the torque imparted by the wind. For a given amount of power, the lower the angular velocity, the higher the torque, and conversely, the higher the angular velocity, the lower the torque. Conventional electrical generators run at speeds many times greater than most wind turbine rotors, so they generally require a gearing system. Wind turbines with few blades (e.g., two or three) are more suited to electricity generation because they operate at high TSRs, and thus do not require high gear ratios to match the speed of their rotors to that of the generator [4].

8.8.2 Maximum Theoretical Efficiency of Wind Turbines

Energy in the wind is a low-quality energy due to the randomness of its distribution. Whenever we convert a low-quality energy into a high-quality energy, such as converting wind energy to a rotational mechanical energy, there will always be a fundamental physical limit on the maximum theoretical efficiency of such a conversion. In the wind-to-mechanical energy conversion, the limit has been derived by German physicist Albert Betz in 1919 and is known by his name as the Betz limit. Let us now derive this limit using a simplified approach. As shown in Figure 8.8, wind approaches (upwind) the turbine with its undisturbed speed v(m/s). This is the same speed at some distance far away from the turbine (i.e., unaffected by the presence of the blades). As the wind hits the blades its speed is reduced by the blades to v_b(m/s) because the blades have extracted some of its kinetic energy. The reduction of the wind speed results in a reduction of its pressure; hence, it expands resulting in the tapered stream tube. As the wind leaves the blades (i.e., the downwind) its speed is v_d (m/s). If the downwind speed is the same as the upwind speed (i.e., if $v = v_d$) then the turbine has not extracted any energy from the wind. If, on the other hand, the turbine extracted all the energy in the wind then the wind speed will be zero at the turbine and therefore no more wind can be allowed to flow (i.e., the power extraction by the turbine will stop). Hence, an ideal turbine must slow the undisturbed wind speed by an optimum amount. In fact, Betz showed that an ideal wind turbine would reduce the wind speed to one-third of its original undisturbed value.

The power extracted by the turbine blades is given by the difference in rate of kinetic energy between the upwind and downwind flows

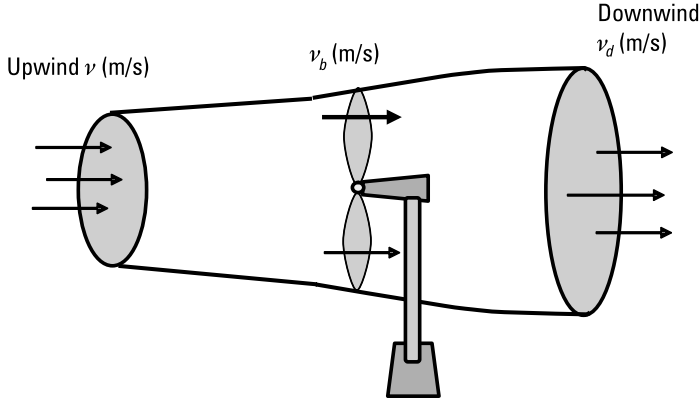


Figure 8.8 Wind approaching the turbine with undisturbed wind speed.

$$P_b = \frac{1}{2}\dot{m}(v^2 - v_d^2) \tag{8.11}$$

This assume that the mass flow rate $\dot{m}$ within the tube is the same everywhere. The mass flow rate at the plane of the rotor, where the swept cross-sectional area is the area of the rotor, is

$$\dot{m} = \rho A v_b \tag{8.12}$$

and

$$P_b = \frac{1}{2}\rho A v_b (v^2 - v_d^2) \tag{8.13}$$

Now we make the assumption that the wind speed at the rotor is the average of the upwind and downwind speeds

$$v_b = \frac{1}{2}(v + v_d) \tag{8.14}$$

Therefore, we can write

$$P_b = \frac{1}{2}\rho A \frac{(v + v_d)}{2}(v^2 - v_d^2) \tag{8.15}$$

Let us define the ratio of the downwind speed to the upwind speed as

$$\lambda = \frac{v_d}{v} \tag{8.16}$$

Hence

$$P_b = \frac{1}{2}\rho A(\frac{v + \lambda v}{2})(v^2 - \lambda^2 v^2) \tag{8.17}$$

Therefore

$$P_b = \frac{1}{2}\rho A v^3 \left[\frac{1}{2}(1+\lambda)(1-\lambda^2)\right] \tag{8.18}$$

The quantity in the square brackets represents the fraction of the power extracted from the wind. This is the power coefficient and is defined as

$$C_p = \left[\frac{1}{2}(1+\lambda)(1-\lambda^2)\right] \tag{8.19}$$

That is, the power available for extraction by the turbine is

$$P_b = \frac{1}{2}\rho A v^3 C_p \tag{8.20}$$

A plot of the variations of the power coefficient C_P with λ is shown in Figure 8.9, which reveals that the power coefficient is maximum when λ is 1/3. This can be verified analytically as follows.

To find when this fraction is maximum, we differentiate with respect to λ:

$$\begin{aligned}\frac{dC_p}{d\lambda} &= \frac{1}{2}\left[(1+\lambda)(-2\lambda)+(1-\lambda^2)\right] \\ &= \frac{1}{2}\left[-3\lambda^2-2\lambda+1\right]\end{aligned} \tag{8.21}$$

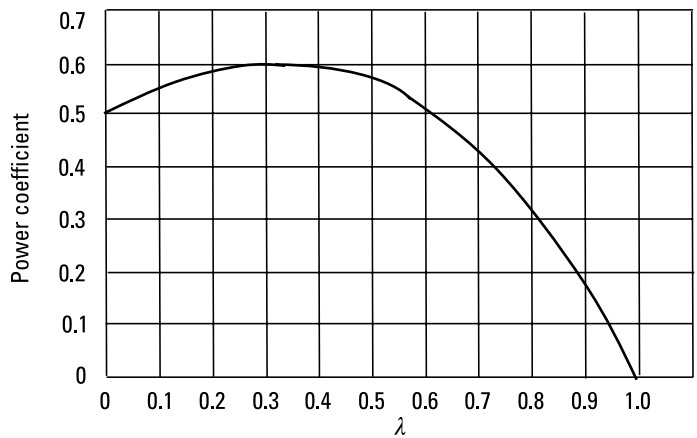


Figure 8.9 The blade efficiency reaches a maximum when the wind is slowed to one-third of its upstream value.

Now we equate this derivative to zero and solve for λ

$$\frac{1}{2}\left[-3\lambda^2 - 2\lambda + 1\right] = 0 \quad or \quad \left[-3\lambda^2 - 2\lambda + 1\right] = 0 \tag{8.22}$$

$$\therefore (1+\lambda)(1-3\lambda) = 0 \quad \therefore \lambda = -1, \ and \ \ \lambda = \frac{1}{3} \tag{8.23}$$

Therefore, the solution is

$$\lambda = \frac{1}{3} \quad i.e.\ v_d = \frac{v}{3} \tag{8.24}$$

That is, the blade efficiency is maximum if it slows the wind speed to one-third of its undisturbed upstream velocity. Now substitute this value of $\lambda = 1/3$ in the equation for the power coefficient (also known as the rotor efficiency)

$$C_p = \left[\frac{1}{2}(1+\lambda)(1-\lambda^2)\right] = \left[\frac{1}{2}(1+\frac{1}{3})(1-\frac{1}{9})\right] = \frac{16}{27} = 0.593 \quad or \quad 59.3\% \tag{8.25}$$

Therefore, we conclude that the maximum theoretical efficiency of a rotor is 59.3%. This is called the Betz efficiency or Betz law. However, wind turbines cannot operate at this theoretical maximum limit, and currently common rotor efficiencies are in the region of 35% to 45%. For a given rotor speed the efficiency of the rotor depends on the speed of the rotor. If the rotor turns very slowly, then efficiency is reduced since the blades are allowing too much wind to pass without extracting any power from it. Similarly, if the rotor runs too fast then the efficiency is again reduced because of the turbulence caused by one blade adversely affects the efficiency of the next one in power extraction. In addition, by the time we consider the other factors in a complete wind turbine system (e.g., the gearbox, bearings, generator, and so on) only 10%–30% of the power of the wind is ever actually converted into usable electricity.

8.9 Electrical Power and Energy from Wind Turbines

The output power of a wind turbine varies, among other things, with wind speed, and every wind turbine has its own characteristic power-speed curve, which is provided in the turbine manufacturer's datasheet. A typical power-

speed curve is shown in Figure 8.10. The shape of a power-speed curve, or simply the power curve, is influenced by several factors, such as rotor swept area, choice of airfoil, number and shape of blades, and the optimum tip-speed ratio. We define the cut-in speed as the speed after which the turbine starts to generate power. Note that although a turbine may start rotating at slightly lower speed than the cut-in, the amount of power it generates may be just enough to supply its losses; hence, there is no point of starting it to simply feed losses. As the speed of the turbine blades increases above the cut-in speed, the output power is proportional to the cube of the speed, and then rises sharply. This continues until the wind speed reaches the rated speed of the turbine and the turbine starts to supply its rated power. If the wind speed is increased any further, the output power is limited to the rated power to prevent damage to the turbine and generator. If the wind speed increases to the shutdown speed, the turbine is automatically stopped by its built-in breaking mechanism.

The energy that a wind turbine will produce depends on both its wind power-speed curve and the wind speed frequency distribution at the site where it is located. The latter is essentially a graph or histogram showing the number of hours in which the wind blows at different wind speeds during a given period. Figure 8.11 shows a typical wind speed frequency distribution. For each incremental wind speed within the operating range of the turbine, the energy produced at that wind speed can be obtained by multiplying the number of hours of its duration by the corresponding turbine power at this wind speed

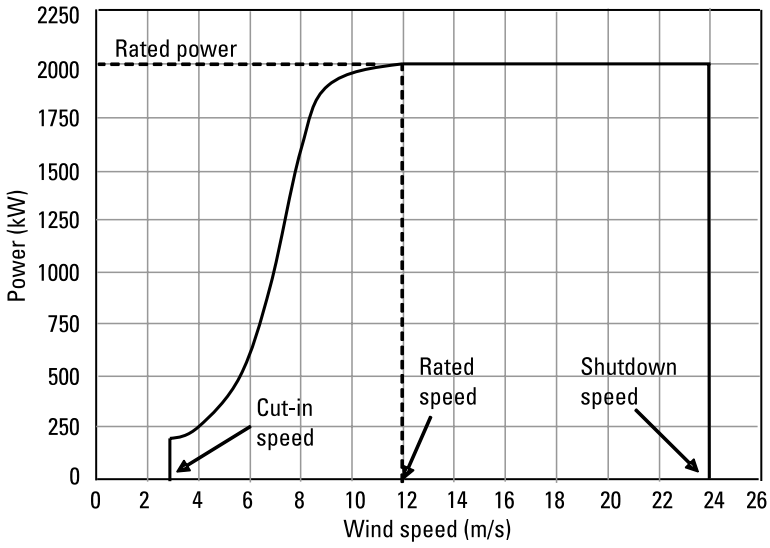


Figure 8.10 Typical wind turbine power-speed curve.

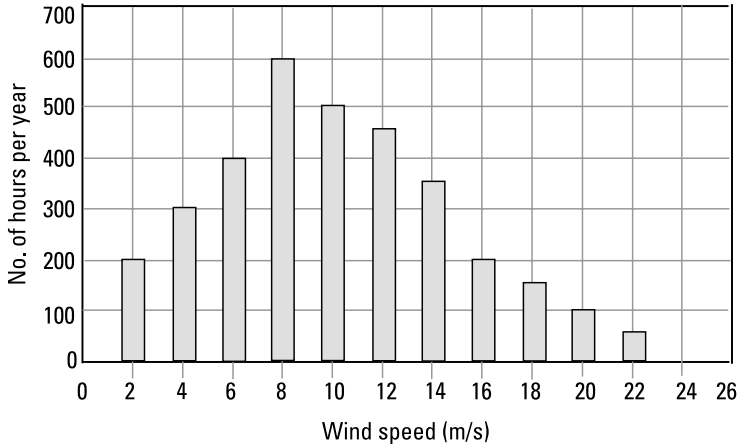


Figure 8.11 A wind speed frequency distribution for a typical site.

given by the turbine's power-speed curve. This data can then be used to plot a wind energy distribution, such as that shown in Figure 8.12.

The total energy produced in a given period is then calculated by summing the energy produced at all the wind speeds within the operating range of the turbine.

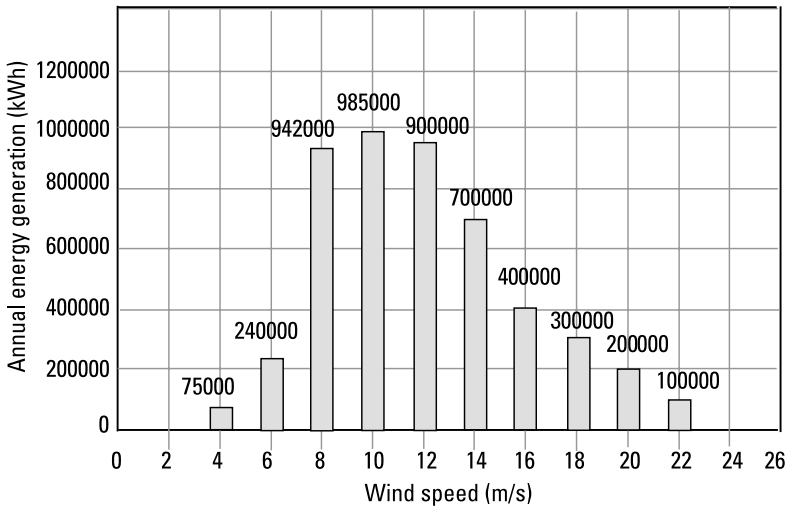


Figure 8.12 Wind energy distribution for the same site as in Figure 8.11, showing energy produced at this site by a wind turbine with the power curve in Figure 8.10.

8.10 Advantages and Disadvantages of Wind Energy

The principle advantage of wind energy is that it uses free fuel (i.e., wind), which is unlimited, and harvesting its kinetic energy does not affect the natural cycle of wind. It is clean and does not produce any greenhouse gases. Disadvantages of wind energy include the relatively high initial costs, particularly for offshore plants. Also, wind turbines can upset wildlife and can be dangerous to birds. Generally, wind turbines must be installed in locations far away from places where the generated energy is needed. This adds to the cost of power transmission and increases losses, which consequently reduces the overall system efficiency. In addition, it increases the cost of maintenance and servicing, specifically for offshore turbine. Finally, wind turbines generate audio and electromagnetic noise, which may not be acceptable to people living nearby and may also harm wildlife. Hence, more research is needed to improve the performance and reduce the cost of wind energy.

8.11 Wind Turbine Generators

Conventional power plants use a synchronous generator, which runs at a fixed speed. Wind energy plants, however, use an induction generator, which is very robust and has simple construction and therefore is more reliable [7]. An induction machine does not require any slip rings or brushes and thus, requires little maintenance compared to the synchronous machine. For wind energy systems, the principle advantage is that an induction machine can run at variable speed as a motor and as a generator. In wind energy systems, an induction machine is used in both modes; as a motor to start spinning the turbine blades and as speed picks up it is switched to generator mode.

8.12 Summary

Wind energy is a renewable source of energy; it is clean and uses wind as fuel, which is sustainable and free. Kinetic energy in a wind stream is harvested and converted to rotational mechanical energy that turns an electric generator using a wind turbine energy conversion system. In this system, there is a theoretical upper bound on the efficiency of the rotor, which is approximately 59%. However, taking losses in various components of the conversion system into account, the overall efficiency is between 10% to 30%. There are two main types of wind turbines: horizontal and vertical axis turbines. HAWTs are the predominant type used in generating electricity. Wind energy conversion systems use induction generators mainly because they are variable speed machines and can run as motors as well as generators.

8.13 Problems

P8.1 a. Write a concise explanation of the statement "wind energy is a form of solar energy".

b. Discuss the advantages and disadvantages of vertical and horizontal wind turbines.

c. Discuss the significance of the blade length and the number of blades in a HAWT paying attention to the output power and application of the turbine.

d. Explain the significance of the power coefficient of a wind turbine.

e. Explain the difference between a drag and a lift type wind turbine.

f. Sketch a typical power-speed curve of a wind turbine and use it to define the following terms: cut-in speed, rated speed, shutdown speed, and rated power of a wind turbine.

g. Define the tip speed and TPR.

h. Discuss two advantages and two disadvantages of wind energy.

i. Explain the dependence of the efficiency of a wind turbine rotor on its speed.

P8.2 Making any reasonable assumptions, derive from first principles an expression for the power delivered to a wind turbine from the kinetic energy of wind.

P8.3 Making any reasonable assumptions, show that the upper bound on the efficiency of a wind turbine rotor is 16/27.

P8.4 A HAWT has a blade length $l = 35\text{m}$ and power coefficient $C_\text{P} = 0.4$. Determine the available rotational power from the turbine when the wind speed is 12m per second and the air density is 1.25 kg per cubic meter.

P8.5 A wind turbine has a diameter of 50 meters. If the wind speed is 12 m/s and the air density is 1.28 kg per cubic meter, how much power does the wind deliver to this turbine?

P8.6 A wind turbine generates an output power of 120 kW when the wind speed is 10 m/s. The turbine diameter is 30m, and the air density is 1.24 kg per cubic meter. Determine the power delivered by the wind and the efficiency of the system.

P8.7 The power-speed curve for a turbine is shown in Figure 8.13.

a. What is the value of each of the following: the rated power of the turbine, the cut-in speed, and the shutdown speed?

b. What is the output power at speed of 6 and 12m per second? Comment on the answers.

c. What is the output power at speed of 8 and 16m per second? Comment on the answers.

P8.8 A HAWT generates 220 kW when the air density is 1.25 kg per cubic meter and the wind speed is 12 m/s. The length of each blade is 15m.

a. Determine the tip speed and the speed of the turbine in revolutions per minute if the tip speed ratio is 4;

b. Determine the required gear ratio if the generator needs to run at 1500 rpm;

c. Determine the efficiency of the overall system.

P8.9 The turbine whose power-speed curve given in Figure 8.13 is located at a site where the annual wind speed distribution is shown in Figure 8.14. Determine the annual electrical energy generated by the turbine.

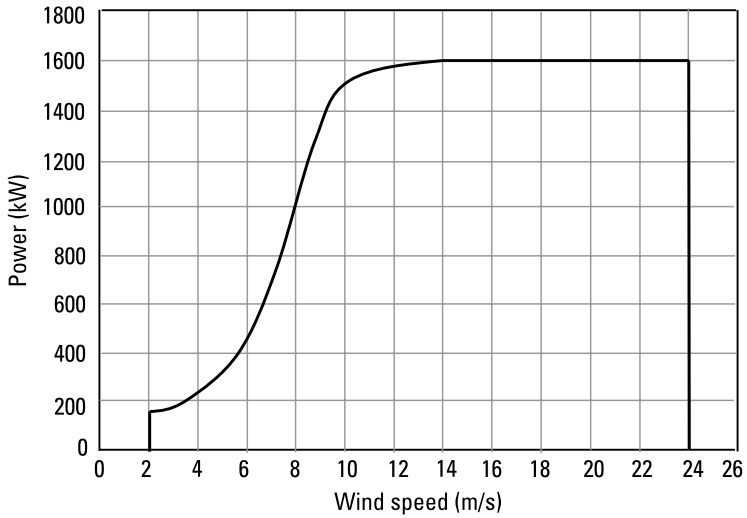


Figure 8.13 Turbine power-speed curve.

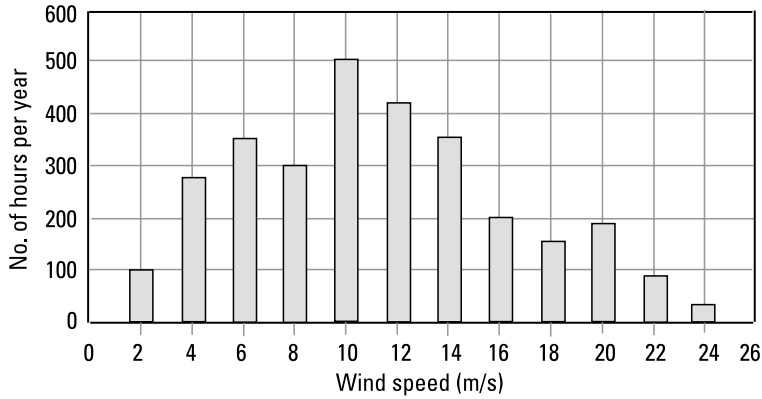


Figure 8.14 A wind speed distribution for a site.

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